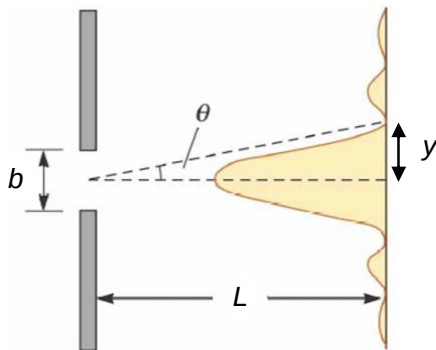


5(a)

Diffraction is the bending or spreading of waves when they travel through a small gap or the spreading of waves around an obstacle.

Note: The common misconception occurs when response displays confusion between gaps and obstacles, e.g. bending of a wave through an obstacle is incorrect.

5(b)(i)



Use $\sin \theta = \frac{\lambda}{b}$, $\tan \theta = \frac{y}{L}$, and small angle approximation, $\sin \theta \approx \tan \theta = \frac{y}{L}$,

$$\frac{\lambda}{b} \approx \frac{y}{L}$$

$$y \approx \frac{\lambda}{b} \times L$$

$$= \frac{590 \times 10^{-9}}{0.100 \times 10^{-3}} \times 2.60$$

$$= 0.01534 \text{ m}$$

The position of the first minima measured from the central axis is 0.01534 m.

No.	Solution
	Width of the central fringe $= 2y$ $= 2 \times 0.01534$ $= 0.03068$ $\approx 0.0307 \text{ m (3 s.f.)}$
5(b)(ii)	Fringe separation, $x = \frac{\lambda D}{a}$ $= \frac{(590 \times 10^{-9})(2.60)}{1.40 \times 10^{-3}}$ $= 1.096 \times 10^{-3} \text{ m}$ Number of fringes resulting from the double-slit interference that are seen within the central maximum produced by single-slit diffraction $= \frac{0.03068}{1.096 \times 10^{-3}}$ $= 27.99$ ≈ 28
6(a)(i)	